

Transcriptomic Re-analysis of Human Liver Tissue Reveals Coordinated Alterations in Protein Synthesis and Protein-Handling Processes in Non-Alcoholic Fatty Liver Disease(NAFLD)

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Abstract

Background

Non-alcoholic fatty liver disease (NAFLD) is a metabolic liver disease associated with obesity, type 2 diabetes mellitus, dyslipidemia, hypertension, and other metabolic disturbances. The disease encompasses a spectrum ranging from non-alcoholic fatty liver (NAFL) to non-alcoholic steatohepatitis (NASH), which may progress to advanced fibrosis, cirrhosis, and hepatocellular carcinoma. At the transcriptomic level, NAFLD is associated with alterations in the gene-expression landscape of human liver tissue. This study characterized these transcriptional alterations and investigated the biological processes associated with them.

Methodology

Publicly available liver transcriptomic data from GSE260666, comprising 16 samples, were re-analyzed using a bulk RNA-seq workflow. Following preprocessing and filtering, 21,141 genes were retained for downstream analysis. Differential expression analysis was performed using DESeq2, followed by Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) over-representation analyses. Preranked Gene Set Enrichment Analysis (GSEA) was subsequently performed to identify coordinated enrichment of biological processes across the ranked transcriptome.

Results

Differential expression analysis identified 64 significant differentially expressed genes (DEGs), comprising 31 upregulated and 33 downregulated genes. Although these DEGs did not show sufficient functional concentration to produce statistically significant GO or KEGG enrichment after multiple-testing correction, transcriptome-wide GSEA identified 494 GO Biological Process terms that were significant at an FDR < 0.05. The strongest positive enrichments included Cytoplasmic Translation (NES = 2.80), Ribosome Biogenesis (NES = 2.63), and Protein Insertion Into ER Membrane (NES = 2.45). Several negatively enriched processes, including xenobiotic metabolism, cholesterol homeostasis, and arachidonic acid metabolism, showed exploratory trends but did not reach the predefined FDR significance threshold.

Conclusion

The enrichment of translation, ribosome biogenesis, and ER membrane protein insertion suggests that NAFLD-associated transcriptional alterations in this dataset involve cellular processes related to protein production and protein handling. These findings may reflect altered proteostatic demands associated with

hepatic metabolic dysfunction and provide a transcriptome-level perspective on the molecular changes accompanying NAFLD. Because the analysis was based on RNA abundance, these findings represent transcriptional associations rather than direct evidence of altered protein synthesis or ER stress.

1. Introduction

Non-alcoholic fatty liver disease (NAFLD) is a systemic metabolic disease driven by interconnected processes involving lipid metabolism, endoplasmic reticulum (ER) stress, inflammation, and fibrogenesis [7]. It predominantly affects individuals with metabolic risk factors such as obesity and type 2 diabetes mellitus (T2DM) [1]. A 2016 meta-analysis of studies published between 1990 and 2015 estimated the global prevalence of NAFLD at approximately 25%, making it one of the most common causes of chronic liver disease worldwide [21].

NAFLD encompasses a spectrum of liver disease characterized by hepatic steatosis in the absence of other secondary causes of hepatic fat accumulation, including excessive alcohol consumption. The spectrum ranges from non-alcoholic fatty liver (NAFL) to non-alcoholic steatohepatitis (NASH), a more progressive form characterized by hepatic steatosis, hepatocellular injury, and inflammation, with varying degrees of fibrosis [2]. Previous transcriptomic analyses of NAFLD and NASH have identified alterations in pathways related to lipid regulation, inflammation, fibrosis, cell death, and survival [5]. Although progression to advanced liver disease may occur over many years, the clinical consequences can include advanced fibrosis, cirrhosis, and hepatocellular carcinoma [22].

At the cellular level, NAFLD is associated with substantial metabolic and organellar disturbances. Patients with NAFLD may exhibit mitochondrial structural abnormalities, reduced respiratory-chain activity, and impaired ATP production [3]. Hepatic steatosis results from an imbalance between lipid acquisition, synthesis, oxidation, and export. Approximately 60% of hepatic triglyceride substrate has been estimated to originate from adipose tissue, 26% from de novo lipogenesis, and 15% from dietary sources [3]. In a population-based cohort of more than four million individuals followed for a median of 4.7 years, overweight and obesity were associated with substantially increased risks of incident NAFLD compared with normal weight [1].

Transcriptomic studies further demonstrate that NAFLD is accompanied by broad changes in hepatic gene expression. A 2019 study reported that liver transcriptomic signatures and quantitative histopathological features were largely similar between NAFL and NASH but differed markedly from healthy normal-weight and obese controls. The most pronounced differences between NAFL and NASH involved pro-inflammatory and fibrogenesis-associated mechanisms [23]. Lipidomic studies in experimental NAFLD models have similarly demonstrated alterations in hepatic lipid composition, including enrichment of triacylglycerols and monounsaturated fatty acids [4].

Although differential expression analysis can identify individual genes whose expression differs between disease and control groups, biological processes are generally coordinated across multiple genes. Consequently, focusing exclusively on individual DEGs may fail to capture broader pathway-level changes, particularly when several genes within the same biological process exhibit modest but coordinated changes. Functional enrichment and gene set enrichment approaches can therefore provide complementary information by examining biological processes at the pathway level [7].

Previous analyses of NAFLD-associated gene clusters have identified pathways involving extracellular-matrix remodeling, inflammation, metabolism, integrin signaling, complement activation, and DNA-damage responses [24]. However, the biological processes identified through transcriptome-wide analysis may vary across cohorts and analytical approaches. Therefore, examining the complete ranked

transcriptome can provide additional information beyond the analysis of genes that meet a predefined differential-expression threshold.

Study Aim

This study aimed to characterize transcriptional alterations associated with NAFLD in human liver tissue by comparing gene-expression profiles between NAFLD and control samples. Specifically, the study sought to identify differentially expressed genes between the two groups and determine the biological processes and molecular pathways associated with these transcriptional changes using Gene Ontology and KEGG over-representation analyses. In addition, preranked Gene Set Enrichment Analysis (GSEA) was performed across the complete transcriptome to identify coordinated pathway-level changes that may not be apparent from analysis of individually significant DEGs. Together, these analyses were used to provide a transcriptome-wide characterization of the biological processes associated with NAFLD in the study dataset.

2. Materials and Methods

2.1 Data Acquisition, Preprocessing, and Normalization

This study utilized publicly available gene-expression data from the Gene Expression Omnibus (GEO; accession GSE260666), comprising transcriptomic profiles from human liver tissue samples. The dataset contained six healthy control samples, six NAFL samples, and four NASH samples, giving a total of 16 samples and 10 NAFLD samples [GEO accession GSE260666].

The count matrix and corresponding sample metadata were processed using Python. Data manipulation, filtering, and numerical processing were performed using pandas and NumPy, while Matplotlib was used for exploratory visualization. The count matrix was inspected for consistency with the sample metadata, and genes with insufficient read counts across samples were filtered before downstream analysis.

Following preprocessing, 21,141 genes were retained for downstream statistical analysis. Count-data normalization was performed within the count-based differential-expression framework to account for differences in sequencing depth and library size between samples.

2.2 Differential Expression Analysis

Differential expression analysis was performed using DESeq2, which models RNA-seq count data using a negative binomial framework and estimates differences in gene expression between the NAFLD and control groups.

Gene-level statistical significance was assessed using model-derived p-values, which were adjusted for multiple testing using the Benjamini-Hochberg procedure. Genes with an adjusted p-value (FDR) < 0.05 and an absolute log₂ fold change ($|\log_2\text{FC}| \geq 1$) were considered significantly differentially expressed.

Log₂ fold change was used to quantify the magnitude and direction of expression changes. Positive log₂FC values represented higher expression in NAFLD samples relative to controls, whereas negative values represented lower expression in NAFLD samples.

Differential expression results were visualized using MA and volcano plots.

2.3 Functional Enrichment Analysis

Functional enrichment analysis was performed using the GSEAPy Python package to investigate biological processes and pathways represented among the significantly upregulated and downregulated genes.

Gene Ontology (GO) Biological Process and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses were performed separately for the upregulated and downregulated gene sets. The background gene set consisted of all 21,141 genes retained for differential expression analysis.

Enrichment significance was assessed using multiple-testing-adjusted p-values, with an FDR < 0.05 used as the significance threshold.

2.4 Pathway Enrichment Using GSEA

To identify coordinated pathway-level changes that may not be captured by analysis of statistically significant genes alone, preranked Gene Set Enrichment Analysis (GSEA) was performed using GSEAPy. All 21,141 genes were ranked according to their differential-expression test statistic and analyzed against the GO Biological Process 2023 gene-set collection.

GSEA was performed using 1,000 permutations. Enrichment was quantified using the normalized enrichment score (NES), while statistical significance was assessed using the false discovery rate (FDR), with FDR < 0.05 considered significant.

Representative enrichment plots were generated for selected significantly enriched biological processes.

3. Results

3.1 Sample and Transcriptome Overview

The GSE260666 dataset comprised 16 human liver samples, including six healthy controls, six NAFL samples, and four NASH samples. Thus, the disease group consisted of 10 NAFLD samples. Following preprocessing and filtering, 21,141 genes were retained for downstream statistical analysis.

3.2 Differential Expression Analysis

Differential expression analysis identified 64 significant DEGs among the 21,141 genes analyzed, comprising 31 upregulated and 33 downregulated genes based on the predefined FDR and $|\log_2FC|$ criteria.

The MA plot showed that most genes were distributed around $\log_2FC \approx 0$, indicating relatively small expression differences for the majority of genes. A subset of genes showed larger positive or negative fold changes. The volcano plot similarly demonstrated that most genes did not meet the predefined significance criteria, while significant genes were distributed on both sides of the fold-change axis.

Among the significant DEGs, TMEM154 showed a positive $\log_2FC$ of approximately 3.54, indicating substantially higher expression in NAFLD samples relative to controls. In contrast, CHRD showed a $\log_2FC$ of approximately -1.45, indicating lower expression in NAFLD samples relative to controls.

3.3 Expression Patterns and Functional Enrichment of Differentially Expressed Genes

The heatmap of representative differentially expressed genes demonstrated distinct expression patterns across the NAFLD and control samples. Expression values were normalized and log-transformed before being converted to per-gene z-scores for visualization.

Over-representation analysis of the upregulated and downregulated DEG sets did not identify GO Biological Process or KEGG pathways that remained statistically significant after multiple-testing correction. Thus, although individual genes showed significant differential expression, the 64-gene DEG set did not exhibit sufficient functional concentration to produce statistically significant pathway enrichment under the predefined FDR threshold.

3.4 Transcriptome-Wide Pathway Enrichment by GSEA

All 21,141 genes were ranked according to their differential-expression test statistic before performing preranked GSEA. The analysis identified 494 GO Biological Process terms that were significantly enriched at $FDR < 0.05$.

The strongest positive enrichments included Cytoplasmic Translation ($NES = 2.80$), Ribosome Biogenesis ($NES = 2.63$), and Protein Insertion Into ER Membrane ($NES = 2.45$). Collectively, these processes indicated a coordinated enrichment of biological functions related to protein production and protein handling.

Several processes showed negative enrichment trends, including Epoxygenase P450 Pathway ($NES \approx -2.03$), Xenobiotic Catabolic Process ($NES \approx -1.94$), Cholesterol Homeostasis ($NES \approx -1.73$), and Arachidonic Acid Metabolic Process ($NES \approx -1.70$). However, none of these negatively enriched processes reached the predefined $FDR < 0.05$ threshold and were therefore considered exploratory trends rather than statistically significant pathway-level findings.

4. Discussion

This study identified transcriptional differences between NAFLD and control liver tissue, with 64 genes showing significant differential expression. Although DEG-based GO and KEGG over-representation analyses did not identify pathways that remained significant after multiple-testing correction, preranked GSEA revealed coordinated enrichment across the transcriptome. The strongest enriched processes involved cytoplasmic translation, ribosome biogenesis, and protein insertion into the ER membrane.

NAFLD is characterized by excessive accumulation of hepatic triglycerides and encompasses a spectrum ranging from simple steatosis to NASH and, in some individuals, advanced fibrosis, cirrhosis, and hepatocellular carcinoma [1]. Although not all patients with NAFL progress to NASH or advanced liver disease, disease severity varies substantially among individuals. Greater steatohepatic activity and increasing fibrosis are associated with a higher risk of disease progression [8].

The strongest finding from the transcriptome-wide enrichment analysis was the coordinated enrichment of processes involved in protein production and protein handling, particularly cytoplasmic translation, ribosome biogenesis, and protein insertion into the ER membrane. Translation and ribosome biogenesis are functionally interconnected because ribosome production and maturation provide the molecular

machinery required for protein synthesis. Previous genome-wide analyses of high-fat-diet-induced liver steatosis have demonstrated substantial changes in translational regulation and shown that transcriptional and translational responses can be independently regulated during fatty liver development, highlighting translation as an important layer of hepatic gene regulation [9].

Similarly, transcriptomic analysis of human liver samples across the progression from healthy obesity to steatosis and NASH identified the ribosome pathway among significantly altered pathways, supporting the relevance of ribosomal biology to NAFLD-associated transcriptional changes [10]. More recently, bioinformatic and experimental analyses of human NAFLD datasets identified ribosomal protein genes as potential NAFLD-associated biomarkers and reported enrichment of the ribosome pathway [11].

The simultaneous enrichment of translation and ribosome biogenesis observed in this study suggests that alterations in the transcriptional machinery supporting protein production constitute a prominent feature of the NAFLD samples analyzed here. However, this finding should not be interpreted as direct evidence that global protein synthesis is increased because RNA-seq measures transcript abundance rather than translation rates. The enrichment of protein insertion into the ER membrane provides an additional connection between protein production and intracellular protein handling. The ER is responsible for the synthesis, folding, processing, and trafficking of many membrane and secretory proteins, and disruption of ER proteostasis has been repeatedly implicated in NAFLD pathogenesis [12–14].

Collectively, the coordinated enrichment of cytoplasmic translation, ribosome biogenesis, and ER membrane protein insertion in this analysis suggests that altered protein-production and protein-handling machinery may represent an important transcriptional feature of NAFLD in this dataset [15,16]. Experimental studies in primary human hepatocytes have further shown that prolonged lipid accumulation can induce ER stress and activate pathways associated with altered protein translation and apoptosis, providing a potential mechanistic link between hepatic lipid overload, impaired protein homeostasis, and hepatocellular injury [17].

The relevance of proteostasis to NAFLD progression is further supported by evidence that ER stress and defective protein-handling responses can interact with autophagy, inflammation, and hepatocyte injury during the transition from steatosis toward NASH [18]. More broadly, metabolic stress in NAFLD involves interconnected disturbances in energy metabolism, ER function, protein folding, and cellular stress signaling [19]. Persistent ER stress and UPR activation may also shift from adaptive responses toward pathways associated with inflammation, hepatocellular dysfunction, and cell death when cellular homeostasis cannot be restored [20].

Taken together, these observations support the biological relevance of the translation-, ribosome-, and ER-related enrichment observed in this study. Rather than identifying an isolated alteration in a single gene or pathway, the present analysis extends previous findings by demonstrating coordinated enrichment of multiple processes involved in protein production and protein handling across the ranked transcriptome. Thus, the analysis provides transcriptome-level evidence that the protein-production and protein-handling axis represents a prominent feature of the NAFLD samples in this dataset.

The absence of significant over-representation analysis results does not necessarily indicate an absence of biological alterations. ORA depends heavily on the genes that cross the predefined DEG threshold, whereas GSEA considers the relative ranking of the complete transcriptome. Consequently, genes that individually exhibit modest expression changes may collectively contribute to the enrichment of a biological process. The present findings therefore demonstrate how pathway-level analysis can reveal coordinated transcriptional changes that are not apparent from a strict DEG list alone.

Several negatively enriched processes, including xenobiotic metabolism and lipid-related processes, also showed exploratory trends. However, none reached statistical significance after FDR correction. These

findings should therefore not be interpreted as definitive evidence of altered cholesterol, fatty-acid, or xenobiotic metabolism in this cohort.

5. Limitations

Several limitations should be considered when interpreting these findings. First, the analysis was based on a relatively small cohort of 16 liver samples, comprising 10 NAFLD and six control samples. The limited sample size may reduce statistical power and may limit the generalizability of the identified transcriptional patterns.

Second, the analysis used bulk liver RNA-seq data. Consequently, the observed expression patterns represent aggregate signals from multiple cell populations within liver tissue. The analysis cannot determine whether individual transcriptional changes originate primarily from hepatocytes, Kupffer cells, hepatic stellate cells, endothelial cells, or other liver cell populations.

Third, this study represents a computational re-analysis of a publicly available dataset and did not include experimental validation. The results therefore provide transcriptomic associations rather than direct evidence of changes in protein abundance, protein synthesis rates, ER stress, UPR activation, or cellular function.

Finally, the findings were derived from a single publicly available cohort. Independent NAFLD datasets and complementary approaches, including single-cell transcriptomics, proteomics, and experimental validation, would be useful for determining the reproducibility, cellular origin, and functional significance of the observed transcriptional patterns.

6. Future Directions

Future studies could validate the identified transcriptional signatures in independent NAFLD cohorts and integrate bulk transcriptomic data with single-cell transcriptomics and proteomic measurements to determine the cellular origin and functional consequences of the observed changes. Experimental investigation of translation-related, ribosomal, and ER-associated processes could further determine whether the transcriptional enrichment observed in this study corresponds to functional alterations in protein synthesis, proteostasis, or ER homeostasis.

7. Conclusion

This re-analysis of publicly available human liver transcriptomic data identified 64 significant differentially expressed genes between NAFLD and control samples. Although these DEGs did not produce statistically significant GO or KEGG enrichment after multiple-testing correction, transcriptome-wide GSEA identified coordinated enrichment of biological processes related to cytoplasmic translation, ribosome biogenesis, and protein insertion into the ER membrane.

These findings suggest that alterations in protein-production and protein-handling machinery represent a prominent transcriptional feature of NAFLD in this dataset. The results also demonstrate the value of transcriptome-wide pathway analysis for identifying coordinated biological patterns that may not be captured by analysis of individually significant DEGs. Further validation using independent cohorts and

complementary molecular approaches will be necessary to determine the cellular and functional significance of these transcriptional changes.

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Figures

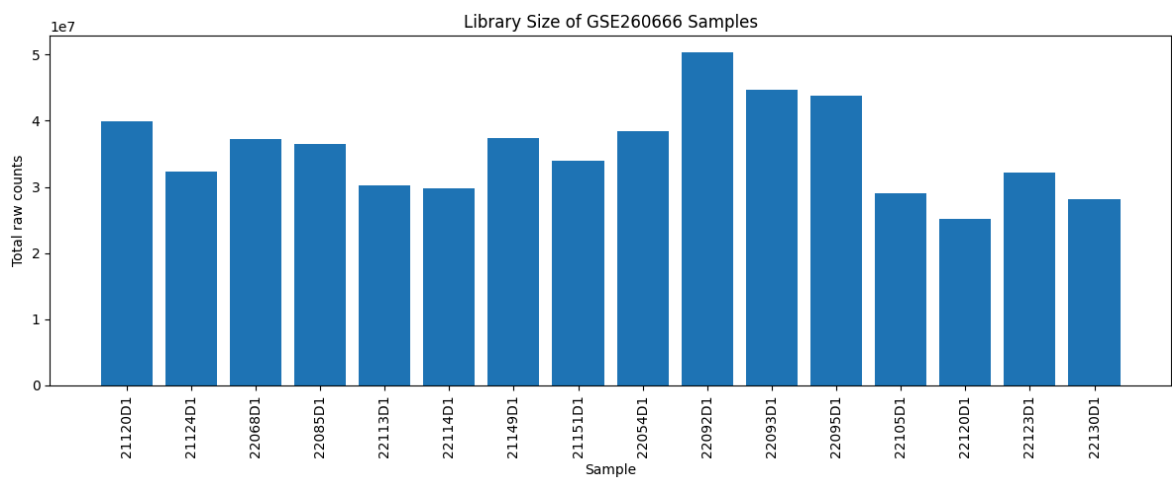


Figure 1

The GEO paper contained 16 samples, with some samples having more raw counts than the rest, but altogether no bizarre raw count differences were observed.

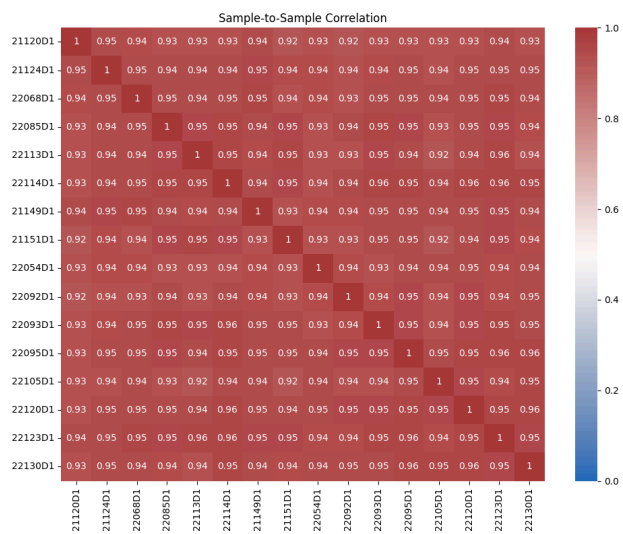


Figure 2

This shows correlations across the 16 samples and indicates that the samples fall within the normal range with level relationships.

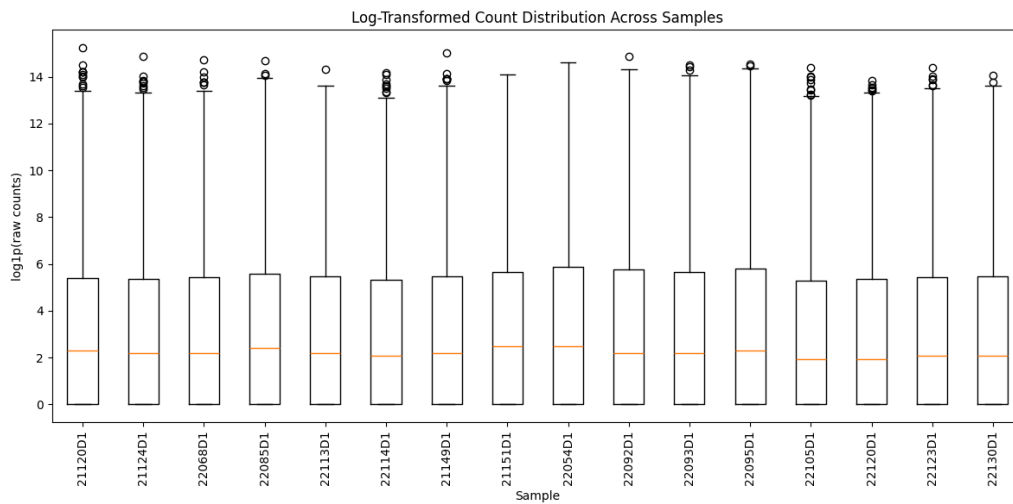


Figure 4

The box plot shows the distribution of log-transformed gene expression values across samples and helps assess whether expression distributions are comparable after normalization and identify potential sample-level abnormalities.

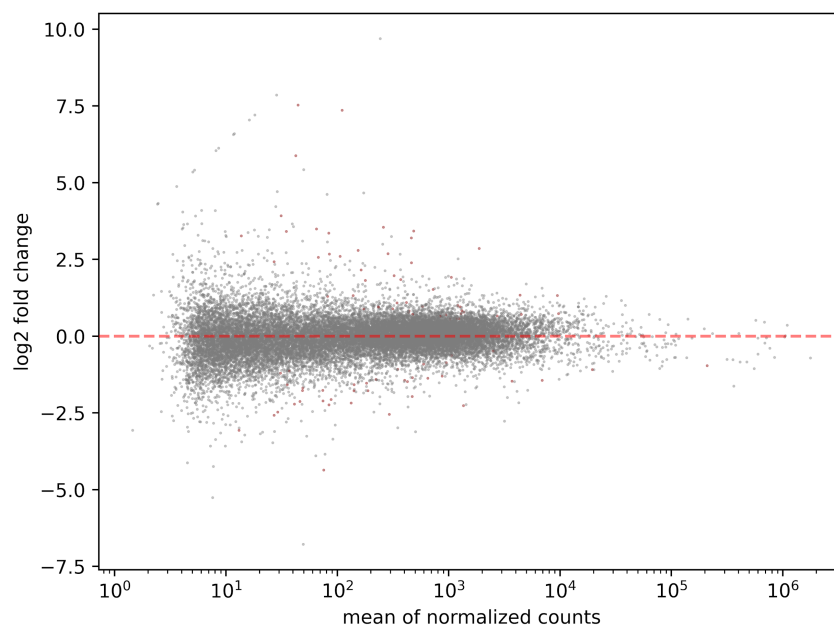


Figure 5

This MA plot shows the relationship between gene expression level and fold change between NAFLD and control samples, which helps to visualize the overall distribution of differential expression and identify genes with large expression changes. Most of the genes are concentrated and fall within the same expression ranges, with very few genes showing large expression changes

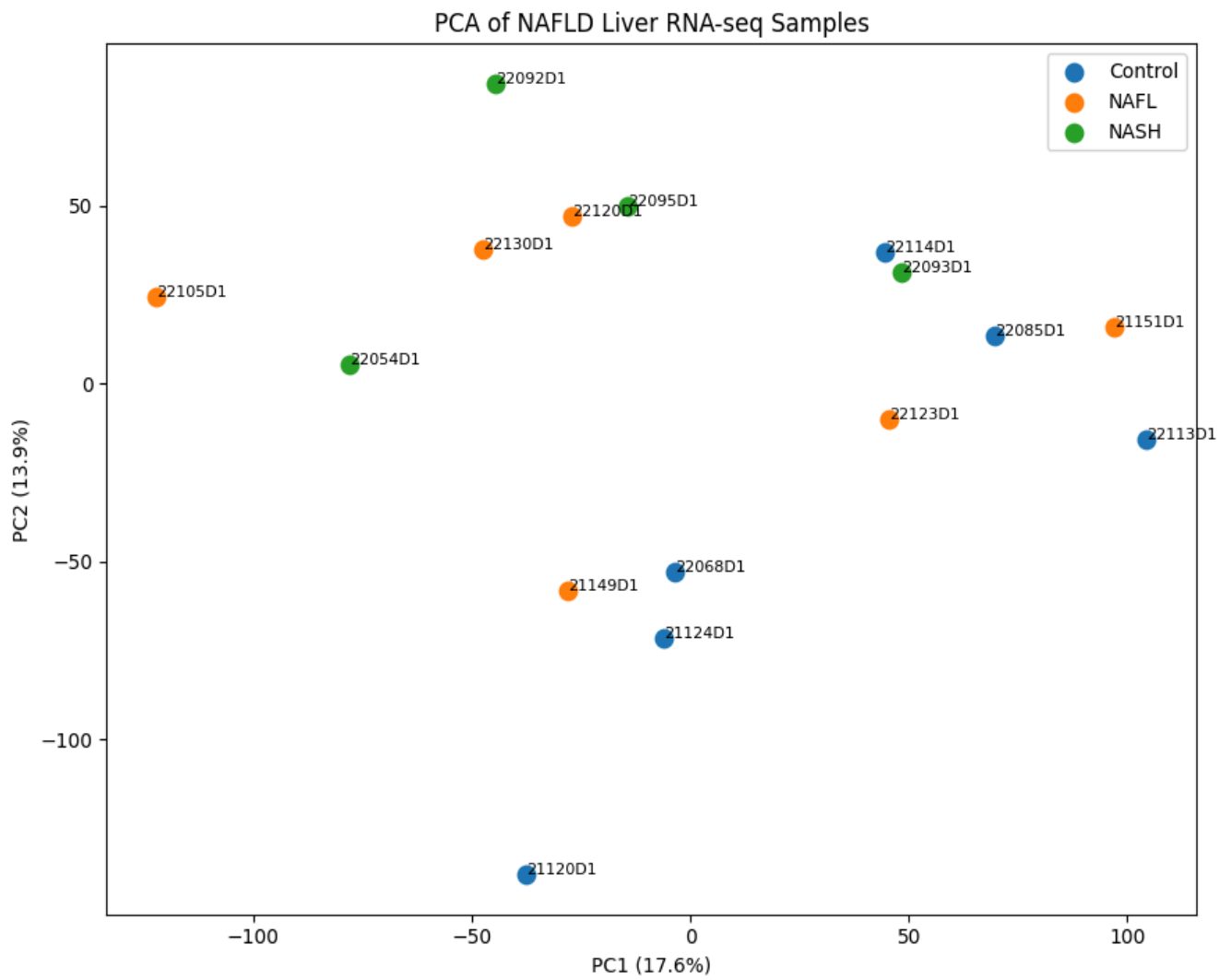


Figure 6

This PCA plot shows the overall similarity and variation among samples based on their gene-expression profiles. It helps determine whether NAFLD and control samples form distinct groups and whether any samples behave as potential outliers.

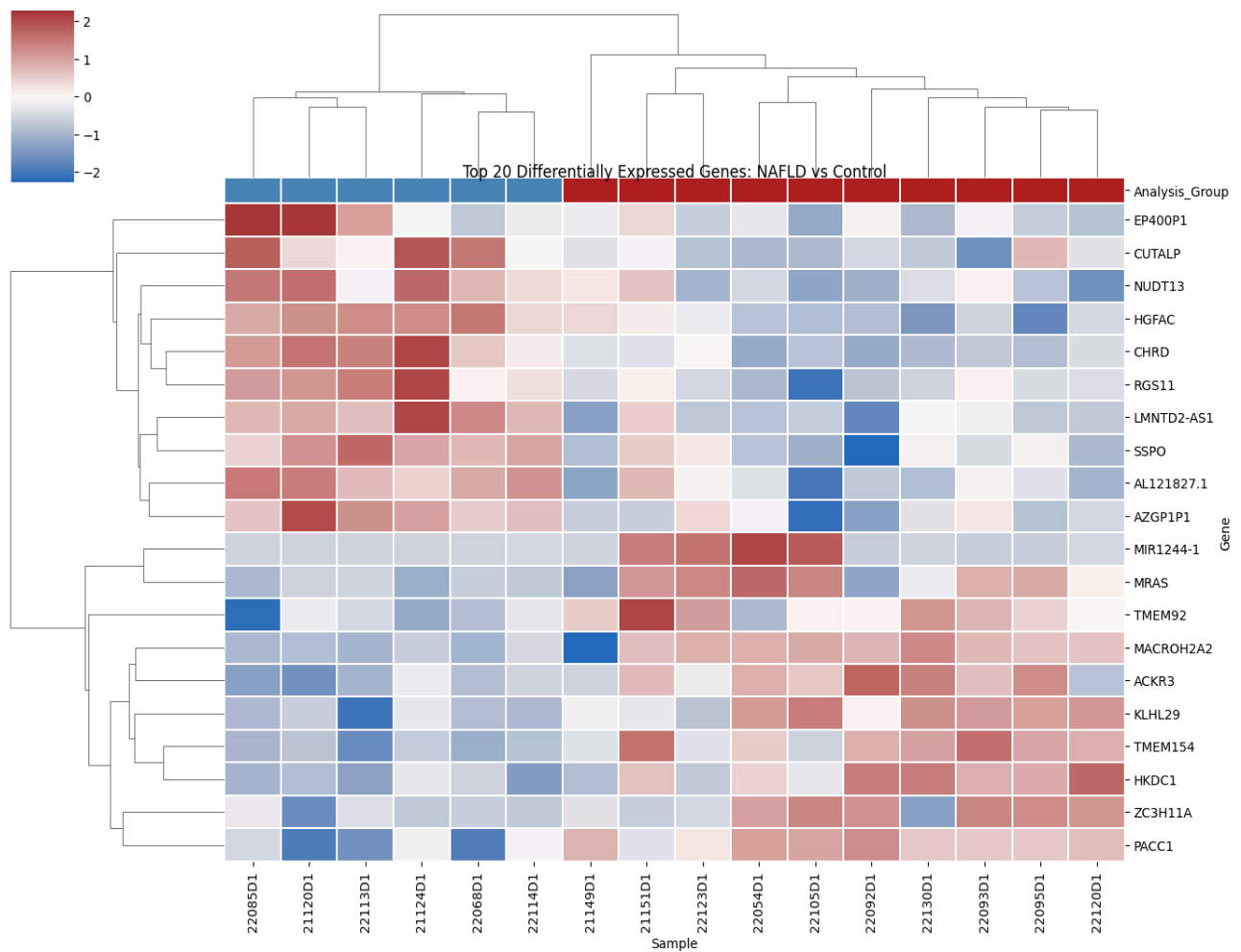


Figure 7

Group heatmap annotation shows the experimental group associated with each sample (NAFLD or Control) alongside the heatmap. We can observe the sample-level gene-expression patterns for the biological groups being compared.

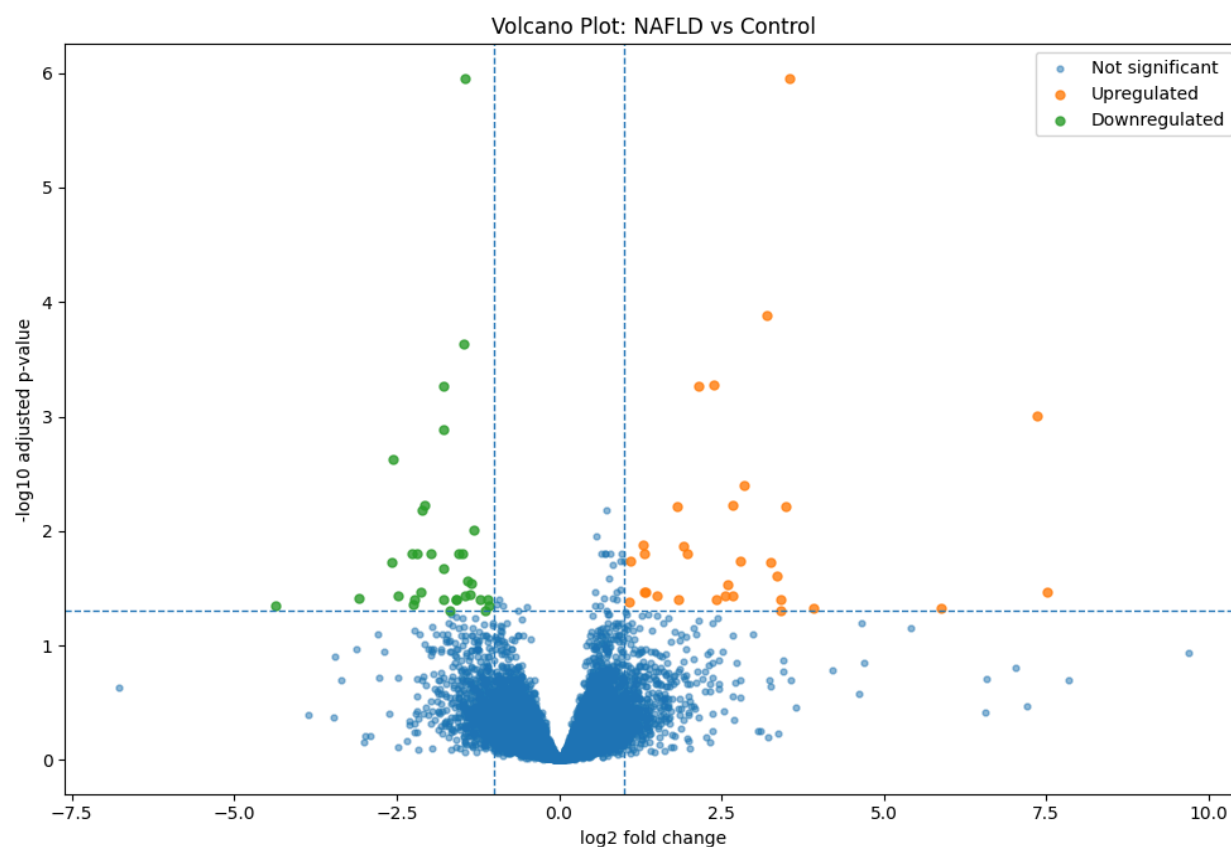


Figure 9

This volcano plot shows the statistical significance and magnitude of differential gene expression between NAFLD and control samples. It highlights genes with both large fold changes and strong statistical evidence, making it easy to identify the most prominent upregulated and downregulated DEGs.

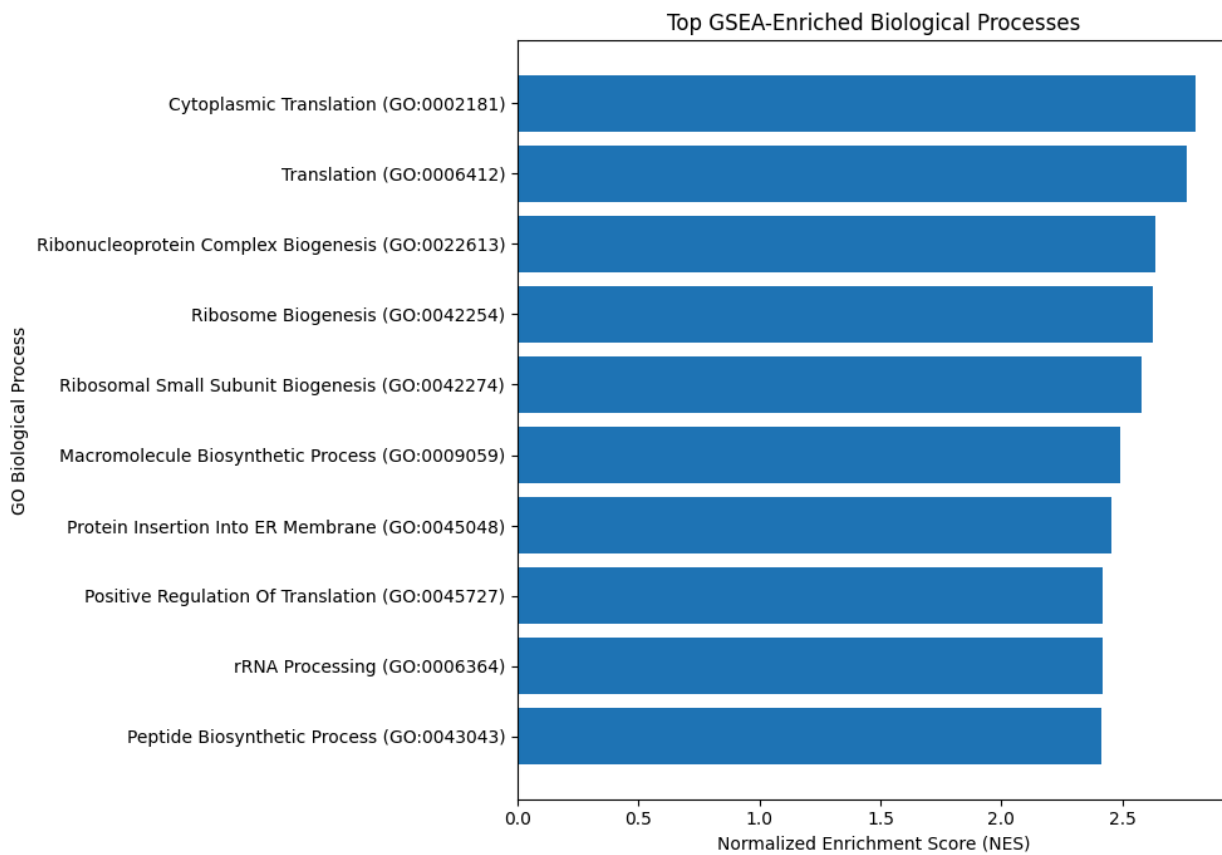
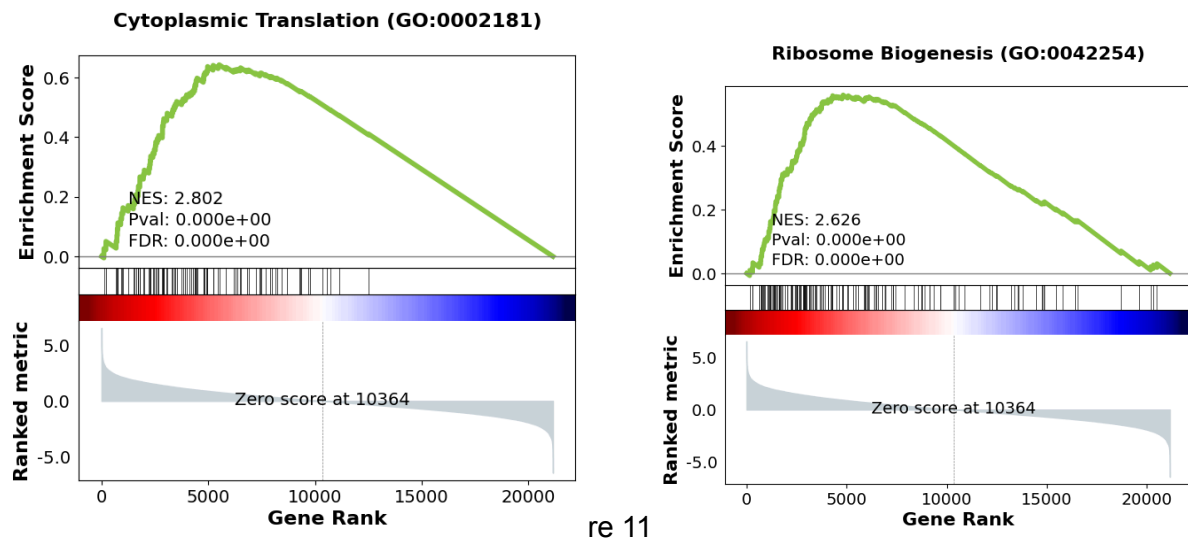
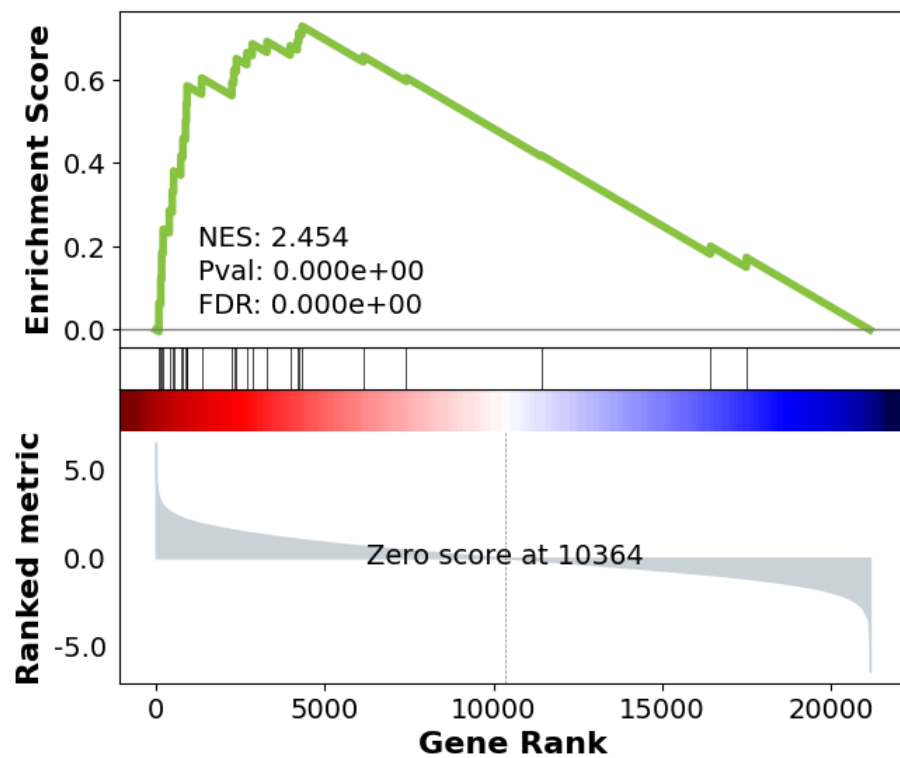


Figure 10

The top GSEA pathway plot shows the most strongly enriched biological processes across the entire ranked gene list, rather than only significant individual genes. It highlights which coordinated gene-expression programs are most associated with the NAFLD samples, such as cytoplasmic translation, ribosome biogenesis, and protein insertion into the ER membrane.



Protein Insertion Into ER Membrane (GO:0045048)



Figures 11, 12, and 13

GSEA enrichment plots showing the distribution of genes within the ranked gene list and their running enrichment scores. The normalized enrichment score (NES) indicates the strength and direction of enrichment, while the nominal p-value and FDR indicate statistical significance after permutation testing and multiple-testing correction, respectively. The zero line represents the point of no enrichment.